

Crime Analysis in Fairfax County

Amber Gamgort | agamgort@gmu.edu

CDS-303 | George Mason University

HOW IS CRIME RATE ASSOCIATED WITH SOCIODEMOGRAPHIC VARIABLES IN FAIRFAX COUNTY?

CONTEXT

This research analyzes how socioeconomic factors influence crime distribution across Fairfax County, VA census tracts. By applying statistical and clustering models, the objective is to identify which socioeconomic variables, such as income, education, and employment, correspond with crime rates to improve public safety understanding.

DATA & PREPARATION

Data was sourced from the Fairfax County Police Open Data Portal (2025–March 2026) and the most recent ACS 5-year estimates (2022) for sociodemographic data. Locations were processed in ArcGIS to allow spatial referencing between datasets. Crime was normalized per 1,000 residents. The final dataset merged ACS and crime data via geospatial connection.

DATA MODELING

Socioeconomic Dimensionality (PCA)

Principal Component Analysis (PCA) was applied to the ACS sociodemographic dataset to identify primary drivers of tract variation and reduce statistical noise. PC1 (29.75% variance) represents a socioeconomic gradient where income and education load positively, while poverty and unemployment load negatively. As shown in the Biplot (Figure 2), the small angles between median income, home value, and rent confirm these variables are highly redundant and act as a single dimension of wealth in Fairfax County. PC2 (22.18% variance) provides further nuance by reflecting specific housing cost dimensions that differentiate tracts with similar overall socioeconomic status.

Cluster Analysis & Crime Correlation (K-means)

K-means clustering grouped census tracts by combining these sociodemographic components with normalized crime rates to identify six distinct clusters, as visualized in Figure 3. This method categorized the county along the established gradient: affluent clusters at the positive end of PC1 represent areas with lower crime risk, while clusters at the negative end highlight economic vulnerability where crime is more prevalent. While the Crime Incident Map (Figure 1) displays the raw spatial distribution of various incident types, the model proves these patterns are fundamentally tied to the integrated socioeconomic landscape defined by the PCA. Normalizing crime data per 1,000 residents was essential to ensure the analysis identified risks based on sociodemographic factors rather than just population density.

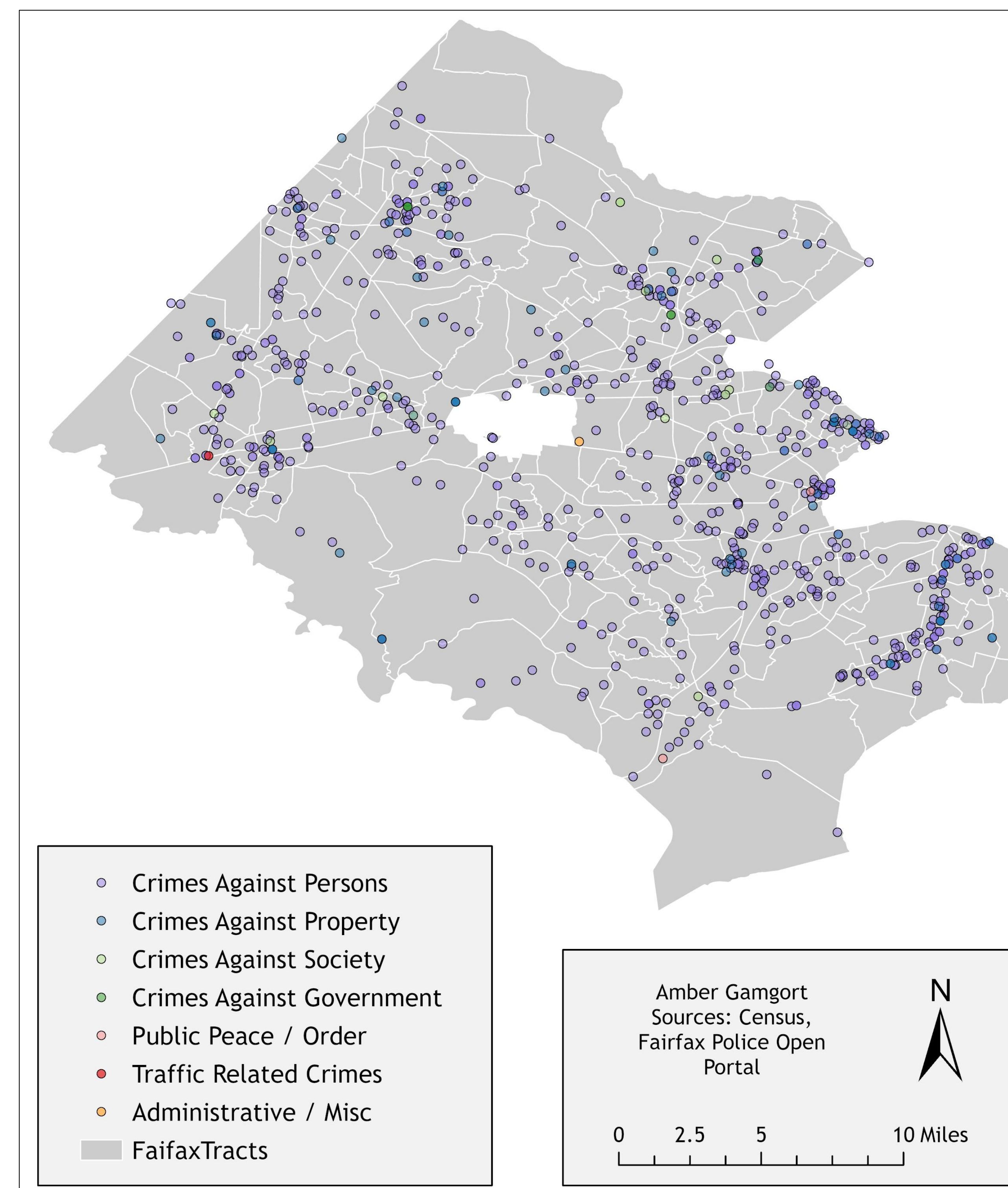


Figure 1. Crime Data Distribution by ID

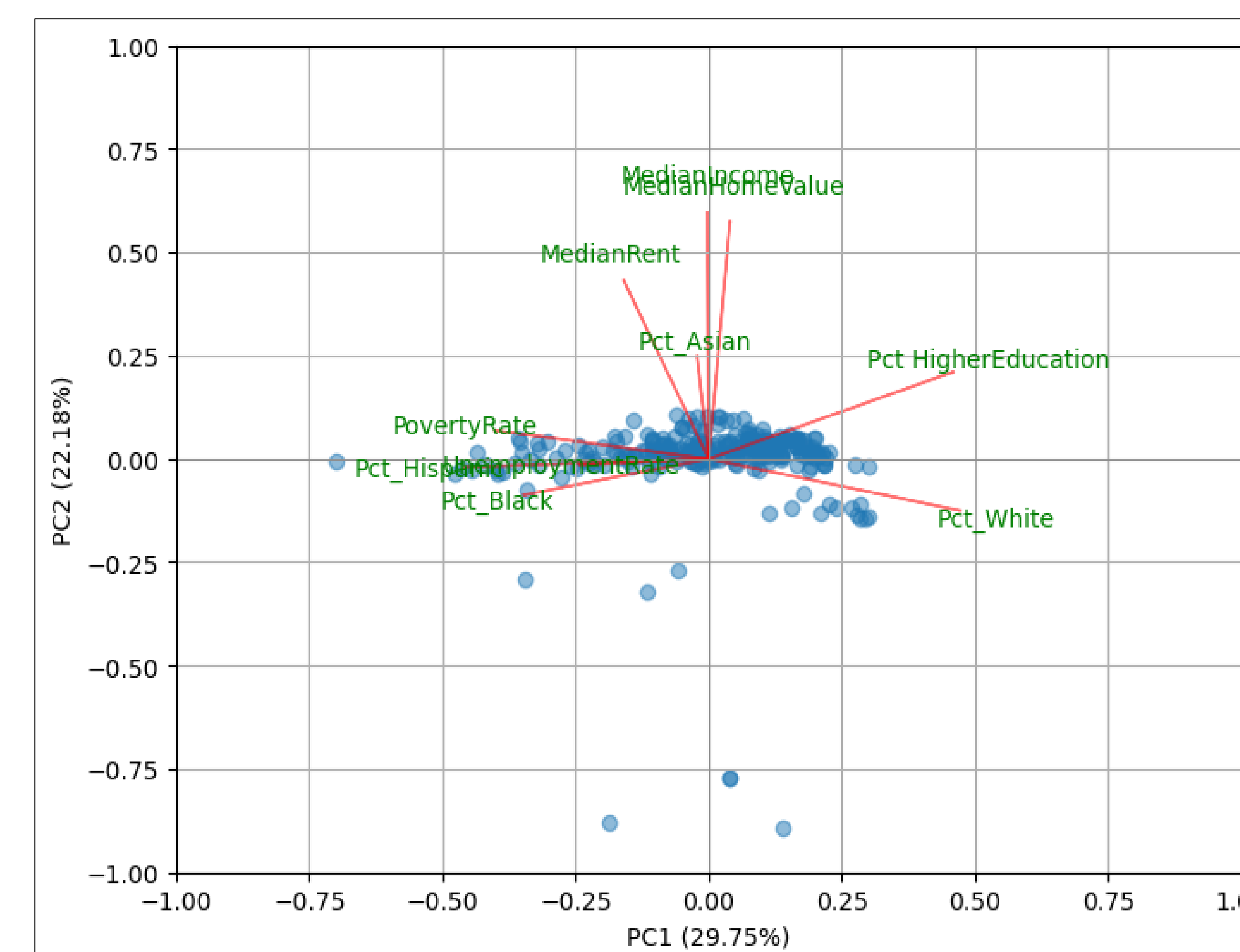


Figure 2. PCA Biplot

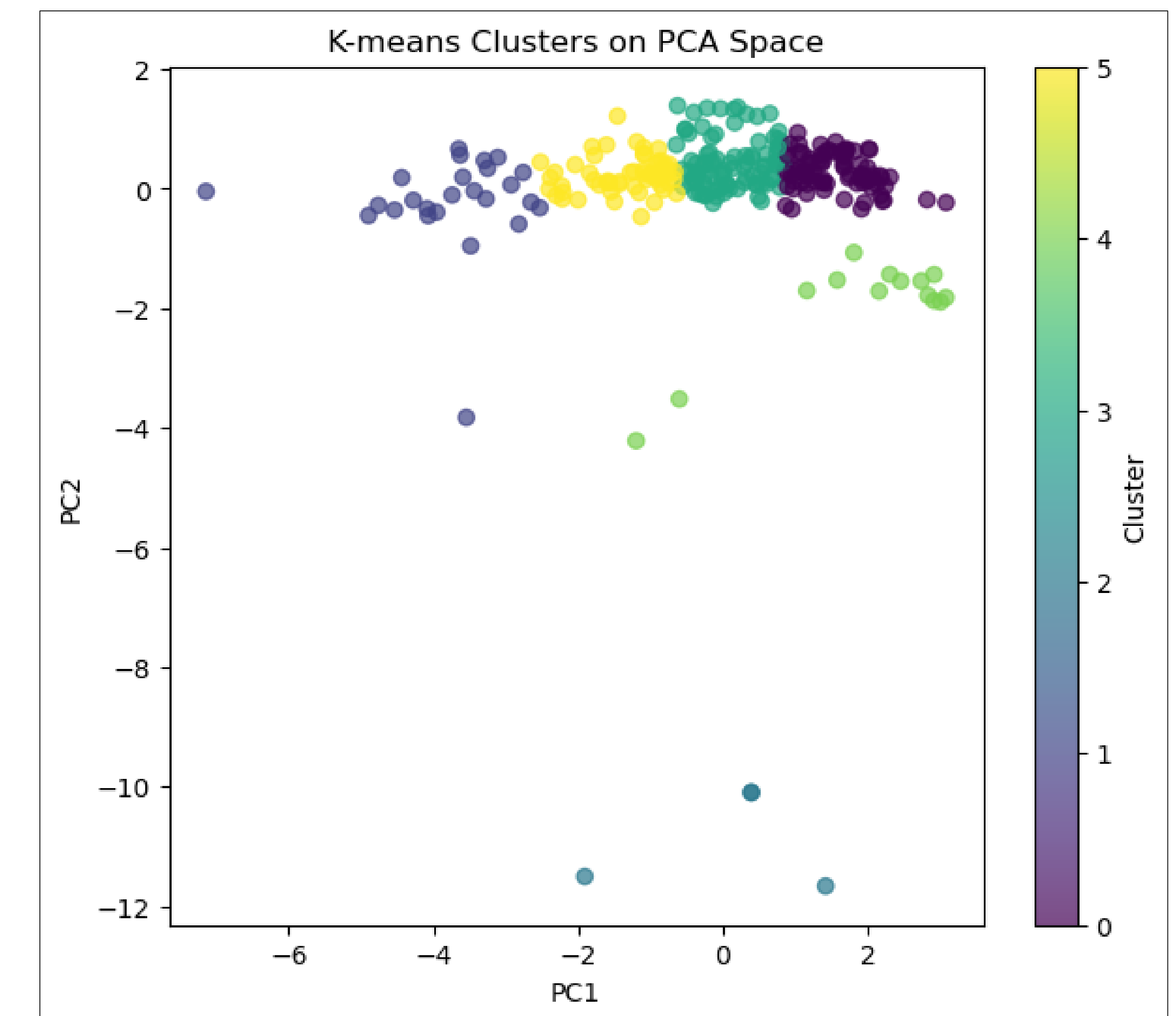


Figure 3. K-Means Clusters

MODEL EVALUATION

The PCA is evaluated by its Explained Variance Ratio, confirming that principal components retain the most significant socioeconomic drivers while reducing data noise. This ensures that factors like income and education remain the dominant variables explaining variation across Fairfax County tracts. The K-means model is validated using the Elbow Method to find the optimal number of groups and Silhouette Scores to ensure clusters are distinct. By then normalizing crime into rates per 1,000 residents, the evaluation confirms that the identified spatial patterns accurately reflect risk relative to population density rather than raw counts.

CONCLUSIONS

Findings suggest implications for urban planning and resource allocation, such as identifying tracts needing increased patrols or improved lighting. This study enhanced my understanding of PCA, clustering, and pattern recognition in large datasets to clarify how socioeconomic conditions impact specific areas in Fairfax County.

REFERENCES:

U.S. Census Bureau. (2022). American Community Survey 5-Year Estimates (2018–2022). Retrieved using Census API from <https://api.census.gov/>
Fairfax County Police Department. (2025–2026). Reported criminal incidents dataset (January 2025–February 2026). Fairfax County Police Open Data Portal.